

Metaheuristic Optimized Machine Learning for Real Time Ammonia Prediction in IoT Based POME Bioremediation Systems

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Abstract— Ammonia removal efficiency in microalgae based Palm Oil Mill Effluent bioremediation systems requires accurate prediction models to ensure optimal process control and environmental protection. Traditional monitoring approaches often lack the predictive capability needed for real-time optimization, potentially compromising treatment effectiveness. An IoT based monitoring was developed and integrated with machine learning algorithms to predict ammonia concentration during *Chlorella vulgaris* cultivation in palm oil mill effluent (POME) media. This study compares and optimizes five machine learning algorithms including Linear Regression, Decision Tree, Random Forest, XGBoost, and LightGBM with the Whale Optimization Algorithm (WOA) and Bayesian Optimization methods. A dataset containing 25,973 minutes of aggregated sensor readings with 99.9% completeness was used for model training and validation via 5-fold cross-validation. XGBoost optimized with the Whale Optimization Algorithm achieved the highest performance. The model demonstrated an R^2 of 0.99, an RMSE of 0.34 ppm, an MAE of 0.14 ppm, and a MAPE of 0.21%. Hyperparameter optimization significantly improved model accuracy compared to baseline configurations, with all optimized models achieving prediction errors below 0.34 ppm. These findings demonstrate the effectiveness of combining IoT technology with metaheuristic-optimized machine learning for accurate ammonia prediction.

Keywords— Ammonia prediction, POME, microalgae, IoT, machine learning

I. INTRODUCTION

POME is produced at a rate of approximately 0.7 to 0.8 cubic meters per ton of fresh fruit bunches processed. POME is considered among the most polluting agricultural wastewaters because of its dark brown color, elevated temperature, and acidic pH of 3.5 [1]. The extreme organic load of the substance is indicated by several parameters. The

biological oxygen demand (BOD) is 25,545 milligrams per liter (mg/L), while the chemical oxygen demand (COD) is 55,775 mg/L. The total suspended solids (TSS) are 18,479 mg/L. The oil and grease concentration is 8,020 mg/L. The total nitrogen content is 711 mg/L, and the ammonia nitrogen concentration is 36 mg/L [2, 3] introduces substantial environmental hazards, encompassing water contamination and oxygen deprivation within aquatic ecosystems. Ammonia is of particular concern due to its toxicity to aquatic life and its potential to cause eutrophication [4]. The unionized form (NH_3) is significantly more toxic than the ionized form (NH_4^+), and toxicity levels are highly dependent on pH and temperature conditions [5, 6]. Untreated discharge has the potential to disrupt ecological balance, inhibit nitrification processes, and threaten communities relying on affected water sources. This underscores the imperative for effective POME treatment technologies with high ammonia removal efficiency for sustainable palm oil production and environmental protection.

Microalgae bioremediation presents a promising alternative for POME treatment, exhibiting the capacity to remove ammonia with efficiencies ranging from 70% to 95%, while concurrently producing valuable biomass through nitrogen assimilation for protein and nucleic acid synthesis [7]. Domesticated *Chlorella vulgaris* has demonstrated ammonia removal rates of 71.4 mg/L/day, achieving near-complete removal of concentrations up to 678 mg/L through gradual adaptation [8]. However, the performance of these systems is governed by complex nonlinear interactions among parameters, including initial POME concentration, pH, temperature, light intensity, photoperiod, and contact time. High POME concentrations have been shown to inhibit growth due to the presence of toxic phenolic compounds [9]. The pH level influences

ammonia speciation and bioavailability. The confluence of the Internet of Things (IoT) and Artificial Intelligence (AI) has precipitated a paradigm shift in the capacity for real-time monitoring of critical parameters. These parameters include pH, temperature, dissolved oxygen, light intensity, and ammonia, along with cloud-based data analysis [10, 11, 12]. This facilitates precise prediction of ammonia removal rates and automated adaptive management. While machine learning models demonstrate potential for predicting bioremediation efficiency, their performance requires optimization through metaheuristic algorithms to avoid poor generalization [13]. This study employs an Internet of Things (IoT) monitoring system with model optimizers to track *Chlorella vulgaris* cultivation in POME [14, 15] to develop a robust predictive system for accurate ammonia forecasting and effective process control in POME bioremediation applications.

The primary contribution of this work is the integration of real-time Internet of Things (IoT) data acquisition with a predictive model for estimating ammonia levels. This study presents an analysis of *Chlorella vulgaris* cultivation in palm oil mill effluent (POME). The Metaheuristic optimizer is employed to refine and enhance the efficacy of machine learning models. Previous research has examined either IoT-based monitoring or machine learning in wastewater treatment independently. In contrast, this work systematically compares multiple machine learning optimizers to identify the most effective configuration for predicting ammonia concentrations. The application of metaheuristic optimization enhances model accuracy and generalization by efficiently searching for optimal hyperparameter combinations, which are often overlooked by conventional methods. The results demonstrate that optimized models outperform standard approaches and provide new insights into the influence of hyperparameters on performance in complex bioremediation systems.

II. LITERATUR REVIEW

A. Microalgae-Cultivated POME for Ammonia Removal

Cultivating microalgae in POME represents a symbiotic approach to wastewater treatment. Microalgae utilize the nutrients present in POME for growth and simultaneously remove pollutants through metabolic processes [16]. The composition of POME, which includes critical nutrients like nitrogen, phosphorus, potassium, various micronutrients, and organic carbon, creates a conducive environment for the growth of microalgae. *Chlorella vulgaris*, a well-studied microalgae in wastewater treatment, can adapt to strong wastewaters. This is because of its strong cell walls, ability to use different energy sources, and efficient nutrient absorption. The microalgae take up ammonia nitrogen in two main ways: by directly turning it into amino acids when nitrogen is limited, and by converting it to organic nitrogen for protein production when there is enough nitrogen [17], [18]. During the process of photosynthesis, microalgae serve to sequester atmospheric carbon dioxide and employ light energy to facilitate metabolic processes.

Chlorella vulgaris grows in POME through several stages: lag, exponential, stationary, and decline. Ammonia removal is most effective during the exponential phase,

when the cells are most active and need the most nutrients. However, raw POME often has a high organic load, dark color, and phenolic compounds, which can block light and slow photosynthesis. Diluting or pre-treating POME can help improve microalgal growth [9]. The chemical form of ammonia is strongly affected by the pH of the cultivation medium. At neutral or slightly alkaline pH, ammonia predominantly exists as ionized ammonium, which exhibits lower toxicity and is more readily assimilated by microalgae. Temperature affects enzymatic activity and metabolic rates, with optimal growth temperatures for *Chlorella vulgaris* typically ranging from 25°C to 30°C. Light intensity and photoperiod influence photosynthetic efficiency and biomass productivity, with light saturation occurring at intensities between 200 and 400 μmol photons per square meter per second for most *Chlorella* species [19].

B. Machine Learning for Ammonia Prediction

Machine learning algorithms are proving useful for predicting ammonia levels in microalgae-based bioremediation systems. They can better capture complex relationships between different operational factors and treatment results than traditional models. In this study, we used seven machine learning algorithms to predict how well *Chlorella vulgaris* removes ammonia during POME treatment. The algorithms included Light GBM, Random Forest, Decision Tree, and XGBoost. We measured model performance using R^2 , RMSE, MAE, MAPE, and MSE.

Linear Regression serves as a fundamental baseline model that establishes linear relationships between input features and ammonia concentration, predicting the output as a linear combination of input variables. Eq. 1 show calculation for linier regression.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \epsilon \quad (1)$$

where y represents the predicted ammonia concentration, β_0 This is the model's baseline, β_i are the feature weights, x_i are the independent variables including pH, temperature, light intensity, POME concentration, cultivation time, and other operational parameters, and ϵ represents the error term.

Decision Tree creates hierarchical rule-based models through recursive binary splitting, evaluating split purity using entropy. Eq. 2 show calculation for decision tree model.

$$Entropy = - \sum_{i=1}^n p_i \log_2(p_i) \quad (2)$$

where p_i is the proportion of samples belonging to class or value range i at a given node, and n is the total number of classes or value ranges. The tree recursively splits data based on features that maximize information gain until stopping criteria are met.

Random Forest constructs an ensemble of decision trees and aggregates their predictions to produce a final output. Eq. 3 show calculation for random forest model.

$$\hat{y}(x) = \frac{1}{B} \sum_{b=1}^B f_b(x) \quad (3)$$

where $\hat{y}(x)$ is predicted ammonia concentration, B is the trees in the forest, and $f_b(x)$ is the prediction from the b -th decision tree. Random forest is constructed using a bootstrapped sample of the dataset, and random feature selection is applied at each node split [20], [21].

LightGBM implements gradient boosting through leaf-wise tree growth strategy, minimizing the objective function. Eq. 4 show calculation for lightGBM model.

$$Obj = \sum_{i=1}^n L(y_i, \hat{y}_i) + \sum_{t=1}^T \Omega(f_t) \quad (4)$$

where $L(y_i, \hat{y}_i)$ is the loss function between actual ammonia concentration y_i and predicted value $\hat{y}_i$, T is the number of boosting iterations, $\Omega(f_t)$ is the regularization term penalizing model complexity at iteration t , and n is the number of training samples.

XGBoost represents an optimized gradient boosting framework building sequential weak learners, with the objective function. Eq. 5 show calculation for XGBoost model.

$$Obj = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (5)$$

where $l(y_i, \hat{y}_i)$ is the loss function quantifying prediction error, K is the total number of trees in the ensemble, $\Omega(f_k)$ is the regularization term for tree k controlling model complexity, y_i is the actual ammonia concentration, and $\hat{y}_i$ is the predicted ammonia concentration [22], [23].

C. Optimization Models for Machine Learning

Selecting appropriate hyperparameters is critical for developing effective machine learning models, as these choices influence accuracy, generalization, and computational efficiency. This study employs two advanced optimization algorithms to fine-tune hyperparameters for ammonia level prediction.

The Whale Optimizer Algorithm (WOA) is selected for its superior exploration-exploitation balance, ensuring rapid convergence while effectively avoiding local optima. Eq. 6 show the position update equation represents the core mechanism.

$$\vec{X}(t+1) = \vec{X}^*(t) - \vec{A} \cdot \vec{D} \quad (6)$$

where $\vec{X}(t+1)$ is the updated position vector representing a new hyperparameter combination at iteration $t+1$, $\vec{X}^*(t)$ is the position solution, $\vec{A}$ is the coefficient vector controlling exploration and exploitation, and $\vec{D}$ is the distance vector between current and best positions [13].

Bayesian optimization is a sophisticated probabilistic approach based on gaussian processes that builds a surrogate model of the objective function mapping hyperparameters to model performance. Eq. 7 show the Expected Improvement acquisition function guides the search process.

$$EI(\theta) = (\mu(\theta) - f(\theta^+) - \xi)\Phi(Z) + \sigma(\theta)\phi(Z) \quad (7)$$

where $EI(\theta)$ is the expected improvement for hyperparameter configuration θ , $\mu(\theta)$ is the posterior mean prediction of model performance at θ from the Gaussian Process, $f(\theta^+)$ is the best observed performance value so far, ξ is an exploration-exploitation trade-off parameter, $\sigma(\theta)$ is the posterior standard deviation representing uncertainty at θ , Φ is distribution function of normal distribution, and Z the standardized improvement [24].

III. PROPOSED METHOD

This study details the data collection procedures and the methodology used to predict ammonia levels during the cultivation of *Chlorella vulgaris* microalgae in POME. Fig. 1 show proposed method.

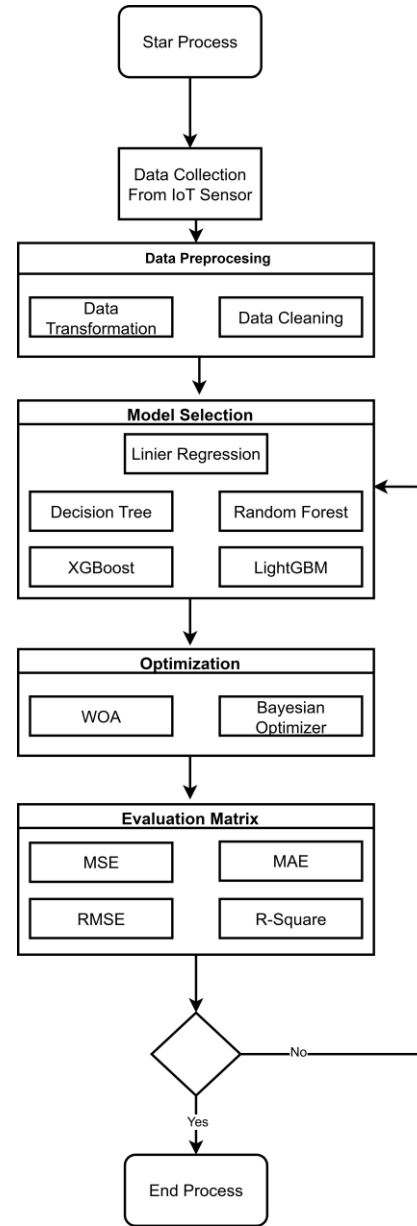


Fig 1. Proposed Method

A. IoT System Settings

The Internet of Things (IoT) monitoring system employs two controllers and designates a Raspberry Pi as the primary data processor to acquire real-time data throughout microalgae cultivation. The initial controller is responsible for the monitoring including pH, turbidity, temperature, total dissolved solids, dissolved oxygen, and ammonia concentration. The secondary controller gauges atmospheric conditions, encompassing light intensity, oxygen concentration, temperature, humidity, and airborne ammonia. Fig. 2 illustrates the system's communication architecture. The system utilizes high-frequency data acquisition, with 10-second sampling intervals, conducted continuously for a period of 18 days. This process enables the capture of the complete *Chlorella vulgaris* growth cycle in POME media. The ammonia values are presented in Fig. 3. The two controllers are configured to transmit raw sensor readings to the Raspberry Pi at a rate of once every 10 seconds. Initially, the data is stored in a local database to ensure data integrity. Subsequently, the data is uploaded to a remote database server via an internet connection, where it is stored permanently. This process enables real-time visualization and facilitates the development of machine learning models.

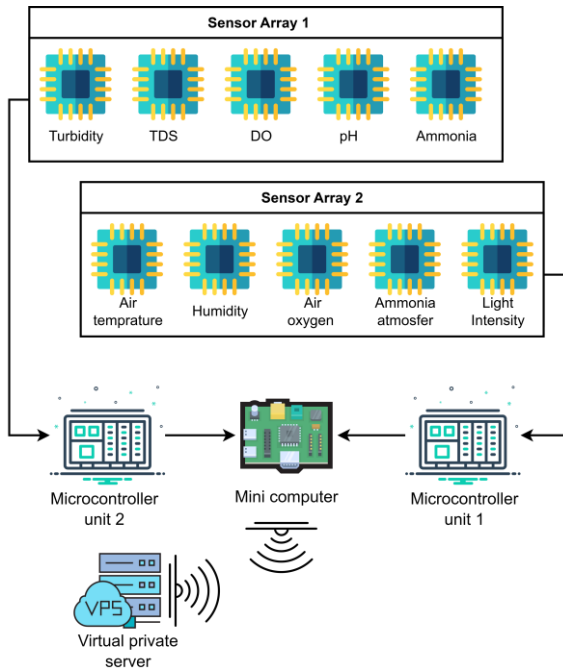


Fig. 2 IoT system Communication

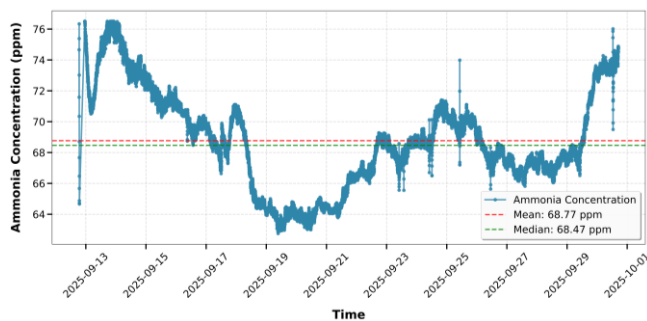


Fig. 3 Ammonia Data from IoT Sensor

B. Cultivation Settings

The microalgae cultivation experiment was conducted from September 12, 2025, to September 30, 2025, with a total duration of 18 days in a controlled bioreactor system. The cultivation medium comprised 600 milliliters of *Chlorella vulgaris* inoculum mixed with 1,900 milliliters of POME at a concentration of 15%, with the addition of 7.5 grams of sodium bicarbonate to ensure optimal pH conditions for microalgal growth. The bioreactor was operated under continuous aeration to ensure adequate oxygen supply and proper mixing of the culture medium. This prevented sedimentation and promoted nutrient distribution throughout the cultivation period. The experiment was conducted under continuous illumination conditions, with an average light intensity of 1000 lux, ensuring optimal conditions for photosynthetic activity and biomass productivity. The cultivation configuration was designed to simulate optimal conditions for ammonia removal while generating the comprehensive dataset required for machine learning model.

C. Model Selection

This research employs five machine learning models to predict ammonia concentration in POME bioremediation: linear regression, decision tree, random forest, LightGBM, and XGBoost. Before modeling, data preprocessing including cleaning and transformation was performed. Model performance is evaluated using R^2 , MAPE, MSE, MAE, and RMSE. Hyperparameter tuning for learning rate, number of estimators, tree depth, minimum samples split, and regularization parameters is conducted using whale optimization algorithm and bayesian optimization.

IV. RESULT

A. Dataset Information

The dataset utilized in this study was acquired through continuous real-time monitoring with a sampling frequency of 10 seconds throughout the 18-day cultivation period, generating a comprehensive raw dataset of 311,660 data points. Table 1 shows an example dataset. To reduce computational complexity and minimize sensor noise while maintaining the temporal resolution needed to capture cultivation dynamics, the raw data were aggregated at one-minute intervals by calculating the mean values of all sensor readings within each minute. The aggregation process yielded an expected total of 26,003 minutes of continuous monitoring. This resulted in a final aggregated dataset comprising 25,973 minutes of valid data, with only 30 minutes of missing data, achieving a data completeness rate of 99.9 percent. The missing data intervals were randomly distributed throughout the monitoring period with no single gap exceeding 5 consecutive minutes, indicating sporadic connectivity issues rather than systematic sensor failure. Quality control procedures included outlier detection using the interquartile range method and cross-validation between redundant sensors to ensure data integrity.

TABLE I. EXAMPLE DATASET

Timestamp	Water Temp. (°C)	TDS (ppm)	DO (mg/L)	pH	Turbidity (NTU)	Ammonia (Water) (ppm)	Light (lux)	Atm. Oxygen (ppm)	Atm. Temp. (°C)	Atm. Humidity (%)	Atm. Ammonia (ppm)
2025-09-12 15:38:00	30.31	2844.42	4.55	8.37	1488.67	108.63	1352.57	268315.81	34.20	51.50	35.92
2025-09-12 15:39:00	30.32	2840.97	4.49	8.38	1490.00	108.67	1289.45	267817.14	34.25	52.22	34.75
2025-09-12 15:40:00	30.34	2841.50	4.50	8.38	1494.00	108.70	1355.97	267963.84	34.30	51.80	37.00
2025-09-12 15:41:00	30.35	2841.18	4.45	8.37	1376.67	108.75	1156.04	268485.88	34.30	51.86	35.00
...	...	...	...	...	...	...	...	...	...	...	...
2025-09-30 16:56:00	36.19	2363.43	6.77	10.00	1158.00	74.79	1357.85	272835.54	38.89	43.56	10.38
2025-09-30 16:57:00	36.19	2367.81	1646.65	10.00	1020.67	74.53	1357.92	272639.06	38.89	43.54	8.67
2025-09-30 16:58:00	36.19	2366.60	6.85	9.95	1107.33	74.77	1358.20	273056.75	38.90	43.50	7.64
2025-09-30 16:59:00	36.19	2367.33	416.64	9.95	1034.67	74.79	1358.41	272964.04	38.90	43.49	10.70

TABLE II. COMPARISON PERFORMANCE OF MACHINE LEARNING MODEL USING HYPERPARAMETER OPTIMIZER

Model	Optimizer	R ²	MSE	MAE	RMSE	MAPE	Best Parameters
Linear Regression	Baseline	0.79	18.81	3.24	4.34	5.32	fit_intercept=TRUE
	BO	0.79	18.81	3.24	4.34	5.32	fit_intercept=TRUE
	WOA	0.79	18.81	3.24	4.34	5.32	fit_intercept=TRUE
Decision Tree	Baseline	0.94	05.07	1.38	2.25	2.11	Max depth=5, min samples split=5
	BO	0.99	0.17	0.13	0.42	0.20	Max depth=25, min samples split=11
	WOA	0.99	0.22	0.14	0.47	0.21	Max depth=29, min samples split=6
Random Forest	Baseline	0.95	0.61	0.42	0.78	0.64	n_estimators=50, max depth=10, min samples split=5
	BO	0.99	0.12	0.11	0.35	0.17	n_estimators=194, max depth=29, min samples split=3
	WOA	0.99	0.12	0.11	0.34	0.17	n_estimators=296, max depth=25, min samples split=2
XGBoost	Baseline	0.98	1.81	0.89	1.34	1.38	Learning rate=0.1, n_estimators=50, max depth=3
	BO	0.99	0.17	0.14	0.41	0.21	Learning rate=0.1919, n_estimators=119, max depth=9
	WOA	0.99	0.12	0.14	0.34	0.21	Learning rate=0.1818, n_estimators=197, max depth=6
LightGBM	Baseline	0.98	2.10	1.11	1.45	1.83	Learning rate=0.05, n_estimators=50, max depth=5
	BO	0.99	0.15	0.19	0.39	0.29	Learning rate=0.248, n_estimators=179, max depth=8
	WOA	0.99	0.14	0.17	0.38	0.26	Learning rate=0.2535, n_estimators=295, max depth=20

B. Regression Result

The study examined five machine learning techniques using default parameters, Bayesian Optimization (BO), and Whale Optimization Algorithm (WOA) for hyperparameter tuning. All optimization procedures used 30 agents, 500 maximum iterations, and early stopping after 10 iterations without improvement. BO employs a probabilistic approach with 70% exploitation probability and 0.1 step size, while WOA uses dynamic coefficient a decreasing from 2 to 0 to transition from exploration to exploitation. Model performance was evaluated using 5-fold cross-validation. The optimized models demonstrated substantial improvements over baseline versions.

Linear regression yielded consistent results, with R^2 of 0.79 and RMSE of 4.34 ppm when `fit_intercept` was set to TRUE. The decision tree model demonstrated improvement, increasing from baseline R^2 of 0.94 and RMSE of 2.25 ppm (max_depth 5, min_samples_split 5) to R^2 of 0.99 and RMSE of 0.42 ppm following Bayesian Optimization (max_depth 25, min_samples_split 11), and RMSE of 0.47 ppm with Whale Optimization Algorithm (max_depth 29, min_samples_split 6). The random forest model achieved baseline R^2 of 0.95 and RMSE of 0.78 ppm using 50 estimators, maximum depth of 10, and minimum samples split of 5. Optimization with Bayesian Optimization increased R^2 to 0.99 and reduced RMSE to 0.35 ppm with 194 estimators, maximum depth of 29, and minimum samples split of 3. Further reduction in RMSE to 0.34 ppm was obtained using Whale Optimization Algorithm with 296 estimators, maximum depth of 25, and minimum samples split of 2. XGBoost improved from baseline R^2 of 0.98 and RMSE of 1.34 ppm (learning_rate 0.1, 50 estimators, max_depth 3) to R^2 of 0.99 and RMSE of 0.41 ppm using Bayesian Optimization (learning_rate 0.1919, 119

estimators, max_depth 9), and RMSE of 0.34 ppm using Whale Optimization Algorithm (learning_rate 0.1818, 197 estimators, max_depth 6). LightGBM advanced from baseline R^2 of 0.98 and RMSE of 1.45 ppm (learning_rate 0.05, 50 estimators, max_depth 5) to R^2 of 0.99 and RMSE of 0.39 ppm using Bayesian Optimization (learning_rate 0.248, 179 estimators, max_depth 8), and RMSE of 0.38 ppm using Whale Optimization Algorithm (learning_rate 0.2535, 295 estimators, max_depth 20).

The XGBoost model optimized with Whale Optimization Algorithm demonstrated the highest performance, achieving R^2 of 0.99, MSE of 0.12 ppm², RMSE of 0.34 ppm, MAE of 0.14 ppm, and MAPE of 0.21%. Random Forest-WOA served as runner-up with R^2 of 0.99, RMSE of 0.34 ppm, MAE of 0.15 ppm, and MAPE of 0.22%, demonstrating comparable accuracy. Both models show superior predictive capability for ammonia concentration in microalgae-based POME bioremediation, suitable for real-time monitoring and control applications.

V. CONCLUSION

Accurately predicting ammonia levels in microalgae-based POME bioremediation is essential for improving treatment efficiency and environmental protection. In this study, we developed an IoT monitoring system integrated with machine learning to estimate ammonia concentrations. We tested five machine learning algorithms, optimizing their hyperparameters with both Bayesian Optimization and Whale Optimization Algorithm. The XGBoost model tuned with WOA achieved superior performance with R^2 of 0.99, RMSE of 0.34 ppm, MAE of 0.14 ppm, and MAPE of 0.21%, meeting the research objective with significant accuracy improvement over baseline models. The developed system demonstrates strong potential for automated ammonia

monitoring and control in industrial POME treatment facilities, enabling precise process management and operational efficiency.

The study acknowledges certain scope boundaries: data were collected from controlled laboratory conditions over a specific experimental period. Future research should validate model performance across diverse operational parameters, seasonal variations, and extended timeframes in pilot-scale or industrial settings. Priority areas include implementing time-series forecasting models for predictive control, integrating automated control systems for dynamic parameter adjustment.

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